# Learning and Knowledge Transfer in Strategic Alliances: A Social Exchange View

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**Abstract**

Although social interactions and exchanges between partners are emphasized as imperative for alliance success, comprehensive examination of how social exchanges facilitate learning and knowledge transfer in strategic alliances is lacking. Drawing on social exchange theory, we examined the effects of social exchange processes between alliance partners on the extent of learning and knowledge transfer in a strategic alliance. An empirical examination of data collected from alliance managers of 144 strategic alliances revealed that social exchanges such as reciprocal commitment, trust, and mutual influence between partners are positively related to learning and knowledge transfer in strategic alliances.

**Keywords:** strategic alliances, interfirm learning, social exchange theory, trust, empirical studies, linear regression

Strategic alliances have become a vital strategy for many corporations to achieve competitive advantage by gaining market access, scale economies, and competence building through collaborations (Dyer and Singh 1998; Gulati et al. 2000; Yoshino and Rangan 1995). Strategic alliances are considered an organizational learning imperative because organizationally embedded knowledge cannot be easily blueprinted and exchanged through market transactions (Doz 1996; Inkpen 1998). Several strategic benefits can be achieved faster, at less cost, with greater flexibility, and with much less risk through alliances than 'going it alone' (Dyer and Singh 1998; Sako 2000). For instance, many organizations are entering into business alliances to overcome the inherent risks associated with new product development, innovation processes, because alliances quicken the speed of innovation, overcome budgetary constraints, and gain access to resources not otherwise available to them (Bleeke and Ernst 1993).

Nevertheless, the actual performance of alliances seems disappointing (Ariño and Doz 2000). Many alliances have been reported to be unstable, ineffective, and poorly performing (Bleeke and Ernst 1993; Geringer and Hebert 1991). The potential for conflict and a clash of interest between alliance partners is inherent, because either party can opportunistically use the alliance to learn the other's business or technological secrets (Doz 1996; Khanna et al. 1998). Since the competitive nature of the process of learning

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and knowledge transfer between firms poses fundamental challenges for academics and practitioners, a new stream of research on how organizations learn from their partners and develop new competencies has emerged.

This new stream of research explores the ecology of learning organizations (Argote et al. 1990), dynamics of learning alliances (characterized as 'learning race') (Hamel 1991; Khanna et al. 1998), evolutionary collaborative processes (Ariño and Doz 2000), and the role of alliance network structure in organizational learning and innovation (Ahuja 2000; Leenders and Gabbay 1999; Gulati et al. 2000). Although this extant research has contributed to the development of certain theoretical premises on the process of knowledge transfer and identified the barriers to interfirm learning, relatively little empirical research integrating the ongoing exchange processes between partners and the learning outcomes has been done. Recently, however, a few pioneering empirical studies have explored the role of relational processes such as trust and conflict management in enhancing interfirm learning (Ariño and de la Torre 1998; Lazaric 1998) and how collaborative know-how and experience, and learning capacity affect the knowledge transfer in alliances (Lane and Lubatkin 1998; Simonin 1997). Since most of the problems in alliances are attributed to the partner firms' inability to learn from each other how to improve their operations through cooperative ways (Hagedoorn and Schakenraad 1994), there is a need to understand how firms can enhance the learning from the alliance, and what can be done to reduce the conflicts and improve the alliance success (Ariño and de la Torre 1998). Consequently, the ongoing dynamic social interactions and exchanges that facilitate interfirm learning assume a greater significance.

In this study, we specifically examine, using a social exchange framework, how social exchanges such as reciprocal commitment, trust, and power-sharing serve as effective collaborative processes in the post-formation phase of alliances and positively influence the learning and knowledge transfer in strategic alliances formed with the intent of learning from their respective alliance partners.

## Learning in Strategic Alliances

Management and organizational theorists have defined organizational learning in a number of different ways, but most common to these definitions is the view that learning involves acquisition and exploitation of new knowledge by the organization (Argyris and Schön 1978; Cohen and Levinthal 1990). Alliances with other firms are often recommended as the means to acquire new knowledge, skills, and expertise to enhance the competency of an organization (Hamel 1991). Despite their advantages, alliances may also introduce the risk of a firm losing its critical capabilities or skills to a partner without receiving any benefits in return. Since either party can opportunistically use the alliance to learn the other's business or technological secrets (Khanna et al. 1998), the potential for conflict between alliance partners is inherent. Such conflicts may render the alliance ineffective and unsuccessful.

The previous alliance research has focused on this issue of partner opportunism and has adopted a transaction cost economics perspective to examine this aspect (Pisano 1989; Williamson 1991). Using transaction cost economics theory, scholars recommend that the high transaction costs resulting from opportunistic behavior can be alleviated through appropriate contractual agreements or equity-based governance structures (Kogut 1988). These agreements or governance structures are expected to create a 'mutual hostage' situation through *ex ante* commitments to an alliance. Several scholars have criticized the transaction cost economics perspective on alliances for its singular focus on partner opportunism and its failure to capture the social exchanges and managerial relationships that exist between partners during the formation and post-formation phases of an alliance.

The extent to which partners can realize their learning objectives is dependent on their absorptive capacities (or receptivity), transparency, and the coordination strategies adopted by partners (Cohen and Levinthal 1990; Lane and Lubatkin 1998; Ring and Van de Ven 1994). Absorptive capacity of a partner depends on the quality of its human resources, knowledge base, resources and organizational culture (Lyles and Salk 1996; Hamel 1991). Absorptive capacity of a partner provides greater leverage because the greater the absorptive capacity of a partner firm, the more knowledge it can appropriate from a given volume of total knowledge generated in the alliance. While absorptive capacity enhances a partner's ability to appropriate the collective knowledge, however, low transparency and non-cooperative behavior of a partner will hinder the interfirm learning. The non-reciprocal intent and unwillingness to share the information by one partner undermines the willingness to cooperate by the other partner. Similar to the prisoner's dilemma of collective action and game theory (Parkhe 1993), narrow self-interest in learning can cause a dysfunctional interorganizational learning and hinder the learning outcomes.

A partner's absorptive capacity is indeed important for effective learning; however, the success of collective learning effort is determined by cooperative social interactions and exchanges between partners. While the individual firm learns by changing its actual routines (Argyris and Schön 1978) or potential repertoire thereof (Hedberg et al. 1976), a strategic alliance of two or more organizations can learn by changing their routines or repertoire of possible joint activities. This calls for an enhanced mutual commitment of resources and joint decision-making.

Also, effective learning and knowledge transfer in an alliance depends on the extent to which partners share knowledge and information. The learning in an alliance is further complicated because often the essential knowledge to be exchanged is tacit (Polanyi 1948) and socially embedded in context-specific relationships (Granovetter 1985), making it difficult to learn. Tacit knowledge is hard to document and not easily transferable, making it difficult to exchange across firms (Inkpen 1998). Even under conditions of transparency, without mutual respect and willingness to accommodate each other's values and beliefs it is difficult for the partners to acquire tacit knowledge. For example, a case study reports how GM found it hard to transfer the manufacturing capability from Toyota without challenging its own fundamental operating philosophies (Inkpen 1998). A harmonious relationship will enhance management effectiveness by reducing conflicts that arise due to structural and cultural differences. A partner who adopts a cooperative mode of behavior is more likely to comply with the rules and discipline necessary to achieve the goals of the alliance. As some recent studies of learning in alliances begin to indicate, the complexities of interfirm learning accentuate the need to consider the dynamic interaction and exchange processes that facilitate this higher-level phenomenon (Doz 1996; Nooteboom 2000; Steensma and Lyles 2000).

In the following section, we elaborate from a social exchange perspective the relational coordination processes and develop the research hypotheses explaining the relationship between social exchanges and interfirm learning outcomes.

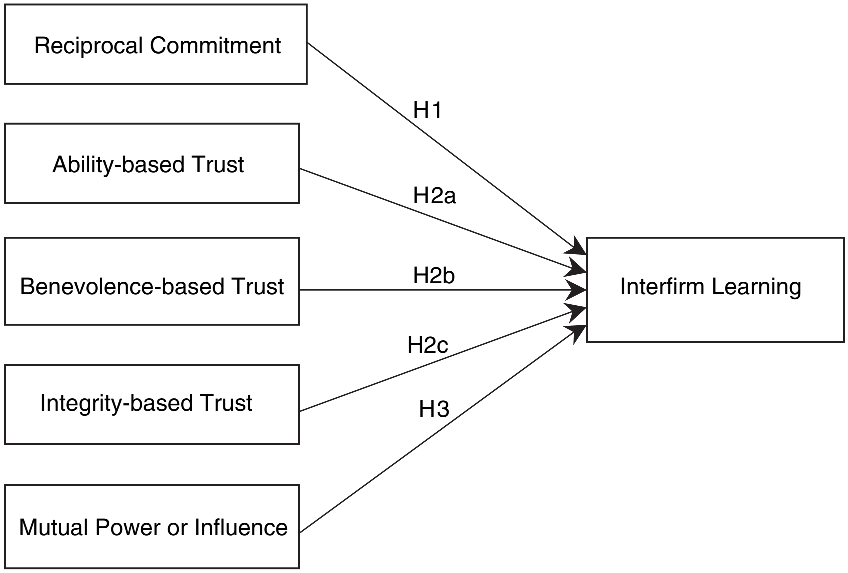
## Theory and Hypotheses

### Alliance Coordination and Social Exchange Theory

According to social exchange theory, 'social exchange' is a situation in which the actions of one person provide the rewards or punishments for the actions of another person and vice versa in repeated interactions (Blau 1964). A mere one-time exchange in a marketplace, where a buyer is able to enter into exchange with one seller on one occasion, with another on another occasion, and so forth depending on the prices offered by sellers, does not qualify as a social relationship. Through repeated interactions, relationships grow, develop, deteriorate, and dissolve as a consequence of an unfolding social exchange and coordination process, which may be conceived of as a bartering of rewards and costs between the partners (Blau 1964; Homans 1961). The benefits of social interaction are intrinsic in nature and have no exact price. Since it is unspecific, it is difficult for the partners to bargain how to reciprocate or force each other to reciprocate. Since there is no way to assure an equivalent return for a favor, social exchange requires trusting others to discharge their obligations. The establishment of friendly partnership requires making investments that constitute commitments to the other party. But it demands trusting others to reciprocate and also proving oneself trustworthy.

Social exchanges are as important in interorganizational relationships as in interpersonal relations (Hosmer 1995; Lane and Bachmann 2000). The various perspectives that emphasize the significance of social exchanges in economic exchanges such as relational contracting (Macneil 1980), social embeddedness (Granovetter 1985), and game-theoretic notions of reciprocal commitment and mutuality are strongly rooted in social exchange theory. The following section reviews the literature on relational social exchanges such as reciprocal commitment, trust, and power-sharing, and their relationship with interfirm learning. A model depicting the relationships between social exchanges and interfirm learning is presented in Figure 1.

Figure 1.
A Model of Relationships between Social Exchanges and Interfirm Learning




**Norms of Reciprocity**

The reciprocity or reciprocal commitment in an exchange manifests in the form of a moral obligation as opposed to a concern for individual gratification (Blau 1964; Homans 1961). This happens because, unlike the economic exchange, social exchange lacks strict accounting (Blau 1964) and the partners are never certain about how much in debt they are to each other, and therefore strong feelings of moral obligation to repay are continually being generated and reinforced. A partner can fulfill this obligation in at least two ways. The first way is increasing the level of inputs to the relationship already being provided. This may increase the feeling in the other partners that they are being out-given, are receiving more than they are giving, or receiving more than they think they deserve (Homans 1961). The second way is to demonstrate their reciprocal gratitude by committing additional resources that would constitute a reward to the party that is over-giving. Such reciprocal behaviors expand the range of resources being exchanged and result in partners acquiring new information, skills, and expertise.

Reciprocal commitment is considered as a sense of duty to the venture and the other partner; it forms the basis on which problems are addressed and solved. Mutuality of commitment reduces the uncertainty for the parties, and enhances the scope for mutual adjustments in the relationship, thus providing a basis for meaningful communication between partners and joint decision-making. Reciprocal commitment of resources by the partner will also enhance the need for joint planning and actions, and result in a high degree of information exchange. When both parties commit their resources they not only learn about each other, but also develop new skills and competencies. This happens because of the complementary resources and information shared by the partners. Reciprocal commitments have a self-amplifying effect on contributions made by individual partners to the alliance because they establish and reinforce the norms that require the other party to join in. Browning et al. (1995) captured in their case study of the SEMATECH consortium how the unconditional contributions by Texas Instruments and Intel in terms of allocating research talents to the venture enlarged the contributions of other firms. Committing time, resources, personnel and physical assets can foster more active involvement between managers at various levels of the organization and their counterparts in the alliance and result in more learning. Reciprocal commitments in terms of personnel and assets enhance the knowledge connections between partners. The knowledge connections facilitate the sharing and communicating of the firm-specific knowledge with partners for creation of new knowledge in the alliance (Inkpen 1998).

*Hypothesis 1: The greater the reciprocal commitment that exists between the alliance partners, the greater will be the degree of learning accomplished.*

### Interfirm Trust

Previous explanations of trust in the interorganizational relational context have revolved around two major concepts: (1) reliance and (2) risk. Thus the trust in an alliance is often defined as reliance on another party under conditions of risk (Nooteboom 1996). The concept of 'reliance' includes two dimensions of trust: confidence or predictability in one's expectations about another's behavior, and confidence in another's fairness or goodwill (Ring and Van de Ven 1992). The inclusion of the 'risk' factor suggests that a party in an alliance would experience negative outcomes from untrustworthy behaviors of the other party (Nooteboom 1996). This condition further means that the greater the risk, the higher the confidence threshold required to engage in trusting action (Browning et al. 1995). Since alliances often involve exchange of technological and proprietary knowledge resources between partners, there is a greater risk of a partner appropriating the resources of the other party in order to eliminate the partner dependence and making the alliance obsolete; there is always a possibility that the resources and investments devoted to the alliance may be of no value and irrecoverable in the event of alliance termination. This study explicitly recognizes the risk factor in the alliance and emphasizes the socio-psychological property of a relationship based on the experience and interaction of alliance managers rather than a conviction or disposition to trust (Ring and Van de Ven 1992; Zaheer et al. 1998). Whereas dispositional trust is an individual trait reflecting expectances that a person would carry from one situation to another (Rotter 1967), a relational form of trust is with specific reference to the partner in the alliance.

As a structural property of alliance relationship, trust is often said to exist between partner firms (Gulati 1995). Firms, however, cannot trust one another since they cannot have attitudes (James et al. 1988). Trust in an interfirm context has often been conceptualized as the agglomeration of trust between individuals in two organizations (Dore 1987; Ring and Van de Ven 1992). As firms work with each other, the individual members of the partnering firms develop trust in their counterparts because of the close relationships and ties that develop between them. It is important, however, to specify what is being trusted. What determines the different levels of trust between partners? In an economic exchange, trust is the positive reputation for non-opportunistic, trustworthy behavior that could reduce transaction cost (Hill 1990). Since this reputation has an economic value, it is economically rational to do exactly what one party has contracted to do (Hill 1990; Hosmer 1995). One approach to explain why a given party will have a greater or lesser amount of trust in another party is to consider attributes of the trustee; that is, the trustor's perception is based on trustworthiness of the particular trustee (Bromily and Cummings 1992; Ring and Van de Ven 1992).

To capture the interfirm trust from the perspective of alliance managers, consistent with the above arguments, a behavioral framework on trust has been adopted in this study. This framework posits that trustworthiness is comprised of three factors: ability, benevolence, and integrity (Mayer et al. 1995). Ability is that group of skills, competencies, and characteristics that are significant to alliance task and allow a partner to have some influence within some operational domains. Thus, ability highlights the task- and situation-specific nature of the trust (Zand 1972). Integrity is defined as the trustor's perception that the trustee adheres to a set of principles that the trustor finds acceptable. Perception of integrity in a relationship is judged by the consistency of the trustee's past actions, the extent to which the trustee's actions are congruent with promises made, and belief that the trustee has a strong sense of justice (Blau 1964; Homans 1961). Benevolence is the perception of a positive orientation of trustee toward the trustor (Mayer et al. 1995). Benevolence-based trust is based on the expectation that another individual or group will not take excessive advantage of the other party even when the opportunity is available, or at least will not knowingly hurt the other's interests (Bromiley and Cummings 1992; Mayer et al. 1995). The conception of benevolent-based trust was apparently the opposite of 'opportunism' or 'self-interest seeking with guile' (Williamson 1985) emphasized in transaction cost literature. Although it is a superlative notion of trust with respect to an interfirm relationship, this argument is consistent with the notion of expectation of fairness and non-maliciousness in a relationship.

Although all three factors are important to trust, each may vary independently of the others (Mayer et al. 1995). These three factors may combine in idiosyncratic ways to reflect various levels of trust in a relationship. In some situations, the trustee's ability may be more important than the other two factors. Other situations may involve simple tasks that do not warrant specific competencies, but the trustor may expect a high degree of integrity from the trustee. When all three factors were perceived to be high, the trustee would be deemed quite trustworthy; in contrast, it is possible for a perceived lack of any of the three factors to undermine trust. However, it is important to examine empirically, in an alliance context, the influence of each dimension of trust on learning and knowledge transfer in an alliance.

The formation of an alliance is based on the acknowledgement that an alliance partner possesses the useful knowledge, experience, and capabilities.

Alliance managers, however, will continue to have confidence in their partners' abilities only if a partner is willing to provide the other party with access to the special knowledge and capabilities that are important to the alliance, and keep the other party interested in the benefits of alliance (Hamel 1991). If a partner is not able to inspire the confidence of the other party with its skill and expertise, this would result in negligence or dominating tendencies that are antithetical to collaborative learning. On the other hand, if the managers do not hold the partner firm in high esteem, they would tend to be pessimistic about the benefits of the alliance and may not be willing to spend their valuable time and effort on the alliance. The trustworthy behaviors of a partner are a precondition for an enriched, meaningful, and continued exchange of knowledge between alliance managers. For instance, Tully (1996) narrates how the alliance between Northwest and KLM Royal Dutch airlines faced dysfunctional conflicts, because both sides thought they knew how to run an airline better than the other partner and wanted to call the shots.

The perceived trust between partners has several other positive consequences. Trust can enhance openness and accessibility toward each other in an alliance. Such openness motivates the partners to be much more transparent, increases the scope of the relationship, and enhances mutual knowledge transfer between firms in the alliance. As the fear of opportunism fades because of mutual trust, coordination and monitoring costs may reduce. On the other hand, lack of trust can destroy the relationship. Doz (1996) reports how the perceived lack of trustworthiness of a partner in a new product development alliance between Alza and Ciba Geigy resulted in conflicts. Alza employees grew suspicious about Ciba Geigy's ulterior motives when the latter was not sharing important information as promised in the agreement. Alza's perception of the integrity and forthrightness of Ciba Geigy deteriorated over time and led to the demise of the alliance.

There are several reports that suggest there is a positive relationship between trust and partner's collaborative behaviors in the form of self-disclosures, information exchange, and cooperative problem-solving (Lazaric 1998; Zand 1972). For instance, Zaheer and Venkatraman (1995) found that trust increased the scope of joint planning and action by partners in strategic alliances. Through relational processes, partners learn about each other's competency and develop confidence in one another. Dore (1987) observed that trust among partner firms in the Japanese textiles industry enhanced the security of the relationship and led to further increase in investments, risk sharing, and knowledge exchange. Thus,

*Hypothesis 2a: The greater the extent of ability-based trust that exists between the alliance partners, the greater will be the degree of learning accomplished.*

*Hypothesis 2b: The greater the extent of benevolence-based trust that exists between the alliance partners, the greater will be the degree of learning accomplished.*

*Hypothesis 2c: The greater the extent of integrity-based trust that exists between the alliance partners, the greater will be the degree of learning accomplished.*

### Mutual Power or Influence

Power in interorganizational relationships refers to the extent of influence that one party has over the other in terms of influencing decision variables that are significant to achieving the objectives of alliance (Yan and Gray 1994). Power relationships may be symmetric (or balanced), where both parties possess the same capability to affect the decisions of the other; but when the power relations are asymmetric (or unbalanced), one of the parties, that is, the stronger party, has greater control or influence than the weaker party. The nature of mutual influence and control is determined by the balance of power in the relationship (Blau 1964).

From a resource-dependence view, it has been argued that interorganizational relationship characterizes a political situation in which resource scarcity and dependence may prompt organizations to exert coercion over the firms that possess scarce resources (Pfeffer and Salanick 1978). However, it must be emphasized that not all dependence-based relations are asymmetric. While power arises from differences in dependencies, all potential power is not necessarily enacted or exercised (Provan and Gassenheimer 1994). Long-term interests and deferred gratification may result in balanced power relations between partners. From a social exchange perspective, however, power is broadly defined as the capability of influencing the other party in exchange transactions through rewarding other party for acceeding to one's wishes (Homans 1961; Blau 1964). By supplying services in demand, one party establishes power over the other party. If these services are not readily obtainable elsewhere, the other party becomes dependent on and obligated to the one providing these services. Providing needed benefits others cannot easily supply is undoubtedly the most prevalent way of attaining power, though not the only one, since it can also be attained by threatening or coercing the other party (Blau 1964).

Coercive power, however, has negative ramifications. If the power demands are too severe, relinquishing the benefits of dependence may be preferable to yielding to the demands of the stronger party (Blau 1964). Asymmetric power in an alliance relationship interferes with joint problem-solving, because the weaker partner may perceive an undue exploitation and may guard against it (Provan and Gassenheimer 1994). Such a defensive mind-set severely taxes the ability of partners to exchange knowledge in unbalanced power contexts. In general, all unbalanced relationships are inherently unstable (Steensma and Lyles 2000). In fact, the restricted use of power may be a fundamental shift in policies of firms entering long-term strategic alliance relationships. If the boundary-spanning alliance managers are willing to restrain the excessive use of power over their partner and at the same time allow the other party to have a say in the operations, then each partner is more likely to have positive feelings or psychological attachment to the relationship.

Power is an important phenomenon underlying procedural justice and distributive justice (Homans 1976). A partner is likely to partake confidently if the other's decisions take into account one's own concerns, ideas, and information, and is likely to view the decision-making as a fair process. On the other hand, an unbalanced relation that neglects the inputs from the other party suffers from negative judgments and psychological detachment. The relative power partners are perceived to possess exerts a strong influence on how they view their contributions as well as distributive outcomes. In unbalanced relations, weaker parties may develop a sense of distributive injustice. The result can be frustration and conflict between partners (Kabanoff 1991). Conversely, partners' mutual influence on each other in the alliance encourages them to engage in democratic and participative processes, and thus facilitates learning and knowledge transfer between alliance partners.

*Hypothesis 3: The greater the extent of mutual influence between alliance partners, the greater will be the degree of learning a partner can accomplish in the alliance.*

## Methods

### Data Collection

This study targeted strategic alliances formed by US business firms (with domestic as well as international business partners) between 1994 and 1998 in the following industrial groups: biotech and pharmaceutical (US Standard Industrial Classification [SIC] code 283); computers and office equipment (SIC 357); software (SIC 737); electronics components (SIC 367); and telecommunications (481). These industries formed the target group, since alliances were most prolific here (Yoshino and Rangan 1995). We believed that the time frame used for selecting the alliances would help capture recent trends in the alliances and allow us to study alliances that were at least a year old and had passed the initial formation phase. This would ensure more consistent research findings regarding post-formation interactions and their effect on the interfirm learning process.

We collected both archival and survey-based information on strategic alliances. The initial search and identification of the target firms and the key informants was conducted using the LEXIS-NEXIS database, Predicast's PROMT business and industry Internet database, the F&S index of corporate change, and other sources such as journals, industry reports, company annual reports, and the web pages of companies. Since the focus of this study was on the dyadic relationships, we screened and omitted complete mergers, acquisitions, and fully owned subsidiaries. A total of 610 alliances formed the final sample for which complete information of the type and scope of alliances and characteristics of partner firms was available. Since the unit of analysis in this study was dyadic strategic alliance as represented by alliance managers, and to be consistent with the recommendations to make use of the most knowledgeable respondents (Bagozzi and Phillips 1982), a self-report questionnaire survey with reference to the specific alliance partner was conducted among key informants — CEOs, vice presidents of alliances, and R&D directors — associated with strategic alliances. The key informants were identified from the news reports, Internet business guides (e.g. http:/biz.yahoo.com), company web pages, and Standard and Poor's *Register of Corporations, Directors and Executives*. Since the central focus of this study is to examine the learning and knowledge transfer in alliances, right from sample design through the questionnaire survey we focused on alliances for which there was an intent of learning expressed in primary and secondary information collected pertaining to these alliances. Most of the alliances examined in this study were primarily R&D, manufacturing, or joint marketing alliances with the core objective of learning and knowledge transfer from their partner in the alliance. Due to limitations of time and access, the study proceeded in two phases. In the first phase, data collection was focused on the responses from one side of the dyad, and sample selection was restricted to US respondents. The second phase of the data collection, focusing on the other side of the dyad, began six months later. During the second phase of data collection, the primary focus was on the alliances for which responses were available from the first phase so that the reliability, validity, and congruence of the information collected could be examined using data from both sides of an alliance.

#### Survey Process

Following Dillman's (1978) recommendations for conducting effective surveys, several steps were taken during the entire survey process: (1) sent letters, or made phone calls before mailing the first wave of surveys; (2) mailed the first wave with a business-reply envelope and detailed letter requesting participation; and (3) mailed the second wave with a reminder letter to all non-respondents, and made follow-up phone calls to about 90 non-respondents during the first phase of data collection. During the second phase of data collection, the researcher made follow-up phone calls to about 35 foreign companies and about 100 US companies. The researcher also used electronic mails to send questionnaires to several managers. The phone calls, however, were productive, and the researcher had the opportunity to interview several alliance executives and received valuable qualitative information regarding the alliance relationship.

### Respondents

Of the 610 questionnaires mailed in the first phase of data collection, 156 (25.57%) responses were received. Twelve responses were unusable because of missing data, resulting in 144 (23.6%) usable responses. This response rate was satisfactory considering similar 15–24% response rates reported in published studies on interfirm alliances (John 1984). Possible non-response bias was examined by comparing the demographic characteristics of survey respondents ($n = 144$) with those of non-respondents and of those returning incomplete and unusable responses ($n = 466$). One-way between-groups analysis of variance (ANOVA) of sales, assets, number of employees, and industry category resulted in a statistically non-significant F of .57 for number of employees ($p = .45$), non-significant F of .96 for sales ($p = .33$), non-significant F of .32 for assets ($p$ = .57), and a non-significant F of .30 for industry ($p$ = .58). Thus, the responding firms did not structurally differ from the non-responding firms in terms of sales, assets, number of employees, and industry category. Based on the reporting of 144 firms, 81 (56.3%) were of non-equity based alliances, 36 (25%) were of minority equity alliances, and 27 (18.8%) were joint ventures. Of the total responses, 95 were domestic partnerships, and 49 were international in scope. The number of employees of the firms responded ranged from 5 to 85,400, and their total sales revenues ranged from $0.58m to $25.30bn.

During the second phase of data collection, only 250 questionnaires were mailed targeting the other side of the alliance dyads, since the primary focus was on 144 alliances for which responses were received in the first phase. The second phase of the survey produced 81 usable responses, of which 36 were from foreign firms, primarily from Europe and Asia. For 69 alliances, data was available from both sides of the alliance. The data from the second phase was primarily used to check the reliability and congruence of the information reported about the alliance. The hypotheses testing and further analysis were performed only with data from the first phase of the survey.

## Measures

This study adapted several multi-item measures developed in previous alliance and interorganizational research in the fields of management and marketing. Face validity of the selected constructs was assessed by two business school faculty members familiar with research in strategic alliances. Before administering the questionnaire, many of the scales were piloted with 20 senior executives of defense equipment manufacturing firms involved in developing a collaborative consortium in Oklahoma. Based on their comments the items were refined. In addition, we conducted in-depth pre-test interviews with two senior alliance executives to refine and re-word the survey items. Factor analysis, correlation analysis, and other statistical methods were used to test and evaluate the measurement-related validity. Details of the constructs and their operationalization are discussed below.

### Interfirm Learning

Interfirm learning is both process and outcome, and an important indicator of alliance success (Simonin 1997). In successful collaborative alliances, there are many learning benefits. First, a partner firm learns about the collaborative process. Powell et al. (1996) argued that interfirm learning involves development of cooperative routines that help manage the interfirm relationships effectively and enable transfer of critical resources. Partnering firms learn to adjust to each other’s concerns and take joint action for co-optation of environment. In the extant research, this has also been referred to as joint planning and action and measured using indicators — that captured the extent of joint planning and forecasting between partners in the alliance (Zaheer and Venkatraman 1995). Firms also absorb specific skills, knowledge and competencies held by the partner. The success of alliances is also reflected by the amount of skills and competencies gained (Simonin 1997). Following the above literature, a four-item measure was designed to capture the learning process as well as the degree of knowledge, skill, or competencies transferred from the partner. All items are seven-point scales ranging from '1 = not at all' to '7 = a great deal'.

#### Reciprocity

From the social exchange theory point of view, reciprocity or reciprocal commitment implies that a partner in the exchange relationship responds to the actions taken by the other in a reciprocal fashion (Blau 1964; Homans 1961). This notion of reciprocity differs from other aspects of social exchanges such as trust. Our focus is to study investments or tangible inputs committed by one party that evoke a moral obligation of the other party to reciprocate the same. This moral obligation of partners is also referred to as a joint input commitment or mutual commitment (Williamson 1991; Gundlach et al. 1995). The literature emphasizes two dimensions in the structure of commitment: credibility, and proportionality or symmetries in commitment of the partners (Gundlach et al. 1995). Credibility refers to the magnitude of the parties' combined commitment. The larger and more significant the resources committed by both partners, the stronger the social norms and processes that enhance collaboration. Following the literature, a six-item measure was developed to capture the extent of resources in terms of finance, technology, physical facilities, managerial resources, and time committed by both parties. To take into account the mutuality, reciprocity was measured as the total sum of a partner's account of the resources committed by itself (three items) and its perception on the extent of resources committed by the other party (three items). All six items are seven-point scales ranging from 'strongly disagree' to 'strongly agree'.

#### Interfirm Trust

This research focuses on the exchange dyad and captures the interfirm trust from the perspective of boundary-spanning alliance managers using a behavioral framework on trust, which explains why a party will have a greater or lesser amount of trust for another party based on the interaction and experience (Ring and Van de Ven 1994). That is, the trustor's perception is based on trustworthiness of the particular trustee (Mayer et al. 1995; Ring and Van de Ven 1992). Following the above conceptualizations, this study adopted a 17-item measurement instrument from Mayer and Davis (1999) that captures the trust in the relationship in terms of ability, benevolence, and integrity of the trustee. 'Ability'-based trust was measured using six items that captured the focal party's perception of the partner's capabilities, knowledge, and skills related to the alliance. 'Benevolence' was measured using five items that captured the extent to which the focal party perceived the partner would not intentionally harm its interests. 'Integrity' was measured with six items that captured the focal party's perception regarding partner's fairness, sense of justice, consistency, and values. The items indicated the extent of the above dimensions on seven-point scales ranging from 'strongly disagree' to 'strongly agree'.

#### Mutual Power or Influence

This measure was developed on the basis of research that addressed interfirm influence and bargaining power (Killing 1983; Provan 1984; Yan and Gray 1994). Four items have been used to measure the relative extent to which a firm can influence the other in decisions concerning marketing, R&D, technology, and finance-related matters, and the extent to which power is balanced between partners. The first and second items measured the influence of the respondent firm over the other firm and attributed the power of the partner firm respectively. The absolute difference between these two scales (the power of the respondent firm and the attributed power of the partner) was used to measure power imbalance and reversed to capture the relative power or influence. This score was used in combination with third and fourth items that capture the extent of mutual influence in the alliance. All items are seven-point scales with 'strongly disagree' and 'strongly agree' as the anchors.

### Control Variables

To extract possible confounding effects of other variables, we included several factors that may have spurious influence on the interfirm learning in alliances. Specifically, the following variables were included: firm size (number of employees), respondent industry, prior alliance experience, alliance type (joint venture, minority, non-equity differences), international scope, and the extent of formal control employed in the alliance. For instance, firm size may influence the learning in an alliance, because large firms may possess slack resources to manage boundary-spanning activities and may be adept at interacting with their partners. Also, the learning may be higher in equity-based alliances than in non-equity alliances, because of the ownership control. Following the generic classification of alliances (Das and Teng 1998; Yoshino and Rangan 1995), dummy variables were assigned by coding 1 for non-equity type alliances, 2 for minority equity alliances, and 3 for joint ventures. Joint ventures are independently incorporated entities that are separated from, but jointly run by, parent firms. Minority equity alliances involve one party taking an equity stake in its partner (or equity swap), but do not involve creation of any separate business entity. The equity stake in such alliances can range from 1% to 49%. Non-equity alliances are contractual agreements without any equity arrangements. Prior business interaction with the partner makes the process of collaboration smooth, and enhances the learning efficiency (Simonin 1997). A dummy variable was coded 1 if the respondent had past experience with the specific partner and coded 0 otherwise. We controlled for the effects of international partnership by including a dummy variable coding 1 if the alliance involved foreign partners and coding 0 otherwise. It is possible that learning may be perceived as more salient in international alliances than domestic alliances, because of the inherent information asymmetry and partners' heavy dependence on their foreign counterparts to achieve strategic objectives. In addition, the survey included four items to capture the extent of reliance on formal and legal controls to monitor the alliance. Since formal and legal controls are often recommended as effective alliance coordination mechanisms, it was considered appropriate to remove their effects from the relationships hypothesized. Following the arguments of Alter and Hage (1993) and Williamson (1991), and the measure of formality in interfirm relationships (Ruekert and Walker 1987), a four-item measure was adapted to capture the extent of formal monitoring and control employed in the alliance. All four items are seven-point scales with 'strongly disagree' and 'strongly agree' as the anchor points.

#### Reliability and Validity

We assessed the reliability of the information from the respondents by comparing the responses from both sides of the dyad ($n$ = 69) on the alliance traits (type and scope of alliance, equity structure, past experience), and interfirm learning items. The correlations between the responses from both sides on these items were significant and ranged from 0.54 to 1, indicating a high degree of reliability of the measures. We also compared the secondary data on alliance type and scope, and firm characteristics with the information collected from respondents. The correlation analysis revealed a high degree of agreement and correlation coefficients ranged from 0.82 to 1, supporting congruence in the data. The validity of the dependent variable 'learning and knowledge transfer' was examined using an archival measure of learning progress developed for a segment of the dataset. We developed an archival measure of 'learning progress' for the pharmaceutical and biotech alliances in the dataset ($n$ = 44) with the help of company reports and Recombinant Capital (www.recap.com), a web portal that provides information on the structure, scope, and progress of the thousands of alliances created in the biotech and pharmaceutical sector. This web portal tracks the status of each alliance and the various stages of its progress in terms of duration, clinical trials, testing, licensing, patent filing, federal approval, marketing, and termination/failure. With the assistance of an MBA student, we coded the extent of progress as a proxy measure of learning in each of the pharmaceutical alliances in the sample with a five-point scale ranging from 'no progress' to 'high progress'. This archival measure was significantly correlated with the survey measure of 'interfirm learning' (r = .73). This archival measure was also significantly correlated with all the independent variables of this study. This further supports the validity of the survey measure. We also performed the regression analysis for the biotech and pharmaceutical sample ($n$ = 44) with an archival measure of learning as a dependent variable. The regression results were consistent with the hypotheses, and supported the theoretical validity of the relationships between the dependent variable 'learning and knowledge transfer', and the predictor variables 'ability-based trust', and 'mutual power between partners'.

The dimensionality and reliability of constructs were examined using correlation analysis, principal components factor analysis, and Cronbach's alpha reliability analysis (Nunnally and Bernstein 1994). The correlation matrix revealed that there was a high degree of association among the items of the respective constructs. A principal components factor analysis with

Table 1. Descriptive Statistics and Correlations

| Variables | Means | s.d. | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 1. Interfirm learning | 4.19 | 1.12 | | | | | | | | | | | |
| 2. Reciprocity | 4.40 | 1.08 | .68 | | | | | | | | | | |
| 3. Ability-based trust | 4.55 | 1.28 | .74 | .73 | | | | | | | | | |
| 4. Benevolence-based trust | 3.76 | 1.21 | .51 | .58 | .63 | | | | | | | | |
| 5. Integrity-based trust | 4.39 | 1.01 | .64 | .73 | .68 | .67 | | | | | | | |
| 6. Mutual Power or influence | 4.73 | 1.10 | .26 | .13 | .24 | .12 | .23 | | | | | | |
| 7. Formal controls | 3.94 | 1.03 | −.20 | −.12 | −.29 | −.35 | −.16 | −.03 | | | | | |
| 8. Past experience | .38 | .49 | .50 | .27 | .40 | .20 | .29 | .10 | −.18 | | | | |
| 9. Alliance type | 1.63 | .78 | .21 | .19 | .20 | .02 | .08 | −.03 | −.06 | .23 | | | |
| 10. International | .34 | .47 | .07 | .05 | .15 | .06 | −.00 | .20 | −.16 | .08 | −.03 | | |
| 11. Firm size (employees) | 4,599 | 11,396 | .15 | .14 | .16 | .12 | .10 | .07 | −.15 | .22 | .31 | .09 | |
| 12. Respondent industry | | | .07 | −.03 | .13 | .17 | −.01 | .17 | −.15 | .00 | .02 | .20 | .17 |

Means, standard deviations, and correlations are based on $n = 144$

All correlations above r = .17 are significant at $p < .05$ (2-tailed test) varimax rotation of all measurement items revealed that the items were loading on to their respective constructs without any cross loadings of the items (with eigen values > 1 and factor loadings of .50 or higher). The factor structure results provided support for the convergent and discriminant validity of the constructs. In addition, unidimensionality of each construct and scale reliabilities were assessed with factor analyses and Cronbach's alpha statistics. Both dependent and independent variables were unidimensional and displayed a high degree of reliability (see Appendix). An inter-item correlation analysis of the 17 items of trust revealed that there is a higher degree of association among items *within* a dimension of trust than *between* dimensions of trust. That is, inter-item correlations within each trust construct are larger than that between constructs. A principal component analysis with 'varimax' rotation was performed on 17 items tied to ability, benevolence, and integrity dimensions of trust. A three-factor model was expected *a priori* as suggested by Mayer and Davis (1999). This analysis produced three factors with eigen values greater than one, which together accounted for 81% of the variance in the data. The measurement items loaded on corresponding single factor levels of 0.50 or higher, with low loadings on other factors. The Cronbach's analysis of reliability of scale confirmed that each dimension of trust displayed a high degree of scale reliability, with alphas greater than .70. Consistent with the recommendations of Nunnally and Bernstein (1994), inter-item correlations exceeded .30 and item-to-total correlations exceeded .50. Since the items displayed a high degree of reliability for each trust dimension, we aggregated the scores of the items for the respective dimension of trust by giving equal weights to each item, and the average score of the items of each construct is used for correlation and regression analyses.

### Analyses

We examined the data with correlation, and regression analyses. Since there was a strong correlation between the many predictor variables, an assessment of multicollinearity was carried out through the computation of tolerance value and variable inflation factor (VIF). The tolerance values were well above the .10 cutoff and the VIFs were much below the 10 cutoff, indicating multicollinearity was not a problem. We also included several control variables to account for the confounding effects of alternative factors that may influence the dependent variables of the study. Since initial regression models did not reveal any relationship between industry categories and the dependent variables, we rank ordered the industry groups on the basis of growth rates and included industry as a single control variable in the regression models.

## Results

Table 1 shows the means, standard deviations, and correlations for all variables. Correlation analysis revealed that most of the correlations between social exchanges and interfirm learning were significant ($p < .01$).

Table 2. Results of Regression Analysis of Interfirm Learning on Social Exchange Processes

| Variables | Perceptual measure of interfirm learning[a] Model 1 | | Archival measure of learning progress[b] Model 2 | |
|---|---|---|---|---|
| | $\beta$ | $\Delta R^2$ | $\beta$ | $\Delta R^2$ |
| **Control variables** | | .27*** | – | .34** |
| Industry | .02 | | – | |
| Firm size (employees) | −.02 | | .11 | |
| Alliance type | .04 | | .12 | |
| International | .03 | | .04 | |
| Past experience with partner | .24*** | | .33* | |
| Formal controls | .00 | | .13 | |
| **Independent variables** | | | | |
| Reciprocity | .26** | .30*** | .12 | .01 |
| Ability-based trust | .34*** | .06*** | .38* | .13** |
| Benevolence-based trust | .01 | .00 | −.02 | .00 |
| Integrity-based trust | .14* | .02** | .11 | .00 |
| Mutual power or influence | .10† | .01† | .42* | .05† |
| $R^2$ | .66 | | .53 | |
| Adjusted $R^2$ | .63 | | .39 | |
| F | 26.12*** | | 3.79*** | |

[a] Regression model with full sample *n* = 144
[b] Regression model with sub-sample of biotech and pharmaceutical alliances, *n* = 44
†*p* < .10, * *p* < .05, ** *p* < .01, *** *p* < .001;
All significance levels are based on 2-tailed tests

The regression models that tested the hypothesized relationships between social exchanges and interfirm learning are summarized in Table 2. The variables were entered in the order presented in the table. First, the control variables, and then the independent variables were entered. The control variables alone explained 24% of the variance (with F = 8.79, $p < .001$; Adj. $R^2 = .24$). There was significant improvement in the variance explained when all the independent variables were entered in the regression ($\Delta R^2 = .39$, with significant F change, $p < .001$). The regression model 1 with both control and all the independent variables was significant (with F = 26.12, $p < .001$). The change in $R^2$ ($\Delta R^2$) statistics suggested that significant variables were stable and did not lose their significance as the independent variables were entered one at a time in the regression analysis. The social exchanges and the set of controlled variables explained 63% of the variance in interfirm learning (Adj.$R^2$ = .63). The regression analysis shows strong support for the hypotheses 1 and 2a, and modest support for hypotheses 2c and 3. The results suggest that after controlling for respondent firm size, industry, prior alliance experience with partner, alliance type (equity/non-equity/joint venture), and the extent of formal control, the reciprocal commitment between partners and trust in the ability and integrity of the partner have a significant and positive relationship with the extent of learning accomplished in the alliance (respective standardized beta coefficients b = .26, $p < .01$; b = .34, $p < .001$, b = .14, $p < .05$). The results, however, do not support hypothesis 2b (positing the relationship between benevolent-based trust and interfirm learning). As the results show, hypothesis 3 (positively linking the mutual power or influence between partners to interfirm learning) was only marginally supported (b = .10, $p < .10$). Overall, the analysis revealed that social exchanges between partners have a significant positive relationship with interfirm learning in strategic alliances.

The regression model 2 reported in Table 2 tested the hypothesized relationships between social exchanges and interfirm learning with an archival measure of 'learning progress' for a sub-sample of biotech and pharmaceutical alliances ($n$ = 44) in the data. The overall model was significant (F = 3.79, $p < .01$; Adj.$R^2$ = .39), significantly supported hypothesis 2a, and marginally supported hypothesis 3. In addition to the hypothesized relationships, regression results indicate that past alliance experience with a partner also significantly related to the extent of learning in a strategic alliance.

## Discussion, Limitations and Implications

The results indicate that relational social exchanges between partners have strong influence on the extent of interfirm learning accomplished in a strategic alliance. The reciprocal commitment between partners is a significant explanatory factor of interfirm learning, as it implies the moral obligation of partners serving as the basis for mutual commitment (Dwyer et al. 1987). As mutual resource commitment increases in the relationship, partnership becomes highly intertwined, leading to a higher degree of interdependence as well as joint coordination. By analyzing the differential effects of interfirm trust, the study revealed the sources of development of trust between partners in an alliance. The regression results indicated that trust in the partner's abilities (ability-based trust) is positively linked to new knowledge, skills, and competencies gained through the alliance. The study also conjectures that partner trustworthiness is central to smooth functioning of the alliance. Without exhibiting social competency in terms of integrity and goodwill, it would be difficult for the partners to have meaningful and productive exchanges of information, knowledge and skills.

In the interorganizational literature, the power construct has long been considered an important factor in the structuring of interfirm relations (Pfeffer and Salanick 1978). From the social exchange point of view, this study captured the relative influence of managers in the alliance relationship. Since power is considered a central property of any relationship (Blau 1964), and most interfirm conflicts occur due to asymmetry in the power of parties in a relationship, mutual power or influence in a relationship is a critical aspect of an alliance. Although there was only marginal support for this variable in this study, the mutual influence of partners on each other for successful collaboration should not be undervalued. Consistent with the notion of social exchange theory, power-sharing is an essential relational coordination process. Future studies need to examine this issue further, with more refined measures. The overall support for the effects of the social exchanges between partners is particularly encouraging given the fact that this study controlled for various alternative explanations of interfirm learning such as past alliance experience, as well as formal controls.

The primary contributions of this research are manifold. First, this study, using the social exchange perspective, captures the interactions between alliance partners during the post-formation phase of the alliances. Consistent with the recommendations of several organizational theorists (March and Simon 1958; Cyert and March 1963), this study helps us understand the behavioral routines and relational processes that are required to enhance interfirm collaborative capabilities. This study also made several refinements to the measurement of constructs. Measures were modified to reflect the ongoing bilateral exchanges. Reciprocal commitment or reciprocity was measured as the total sum of a respondent's account of the resources committed by itself and its perception on the extent of resources committed by the other party, and the interfirm trust was captured by the perceived trustworthiness of the other party. Similarly, the mutual power (or influence) measure considered the extent of influence each partner exercised over the other in the alliance.

There are several theoretical and empirical limitations to this study. They open up opportunities to extend or modify the scope of this study along several dimensions. The study neglects an important fact: that social exchange processes are shaped by the societal, economic, cultural, and institutional contexts in which firms and managerial actions are embedded and how these forces determine the cooperative behavior as well as outcomes in alliances. Also, the role of national culture, organizational, cultural, and structural factors in determining the learning capacity (absorptive capacity) and learning outcomes would be an interesting research avenue. This research did not address the role of instrumental processes such as socialization and communication processes in interfirm learning. Another related line of research would be to study the role of information technology in interfirm collaboration. Considering the scope of the study and the limitations due to the size of the sample, this study focused primarily on social exchanges. Future studies will examine these issues with additional data.

Empirically, this study conjectures that there are causal and temporal linkages between the relational social exchange processes and interfirm learning. But, with cross-sectional studies, it is difficult to establish causality. Certainly, a longitudinal examination to capture the dynamics of ongoing interactions and alliance outcomes would be a most appropriate way to test and confirm the hypotheses made in the study. Although this study relied on data from one side of the alliance dyad, it must be noted that we improved the reliability of measures by capturing the bilateral nature of the exchange process. We tested the reliability of the measures by collecting an adequate number of responses from both sides. However, an important measurement limitation of this study is the use of single informants, and to improve the validity of organizational level constructs, use of multiple respondents would be more appropriate (Kumar et al. 1993). Nevertheless, the informants of the study were highly knowledgeable about overall corporate strategic activities and the performance implications of specific alliance.

Although the data for this study came from respondents who were closely involved in monitoring and managing alliances, the study did not address the potential problems of common method variance (Campbell and Fiske 1959) or related concerns about the consistency motif and the social desirability bias (Podsakoff and Organ 1986). To test the common method variance, we employed Harman's single-factor test (Harman 1967), a *post hoc* test. The results revealed no single or general factor suggesting that any systematic variance common to the measures was lacking. Finally, we believe that the anonymity and confidentiality maintained throughout the study reduced the effects of social desirability bias (Konrad and Linnehan 1995).

## Appendix: Measurement Items

**Interfirm Reciprocity (6 items, $\alpha$ = .92) (1 = Strongly disagree, 7 = Strongly agree)**
(1) Our firm has committed a substantial amount of financial resources to participate in the alliance with the partner; (2) Our managers have spent a lot of time and energy to maintain the alliance; (3) Our firm has committed substantial human, technological, or marketing resources in the alliance; (4) The partner has committed a substantial amount of financial resources to participate in the alliance with our company; (5) The partner firm's managers have spent a lot of time and energy to maintain this alliance; (6) The partner firm has committed substantial human, technological, or marketing resources in the alliance.

**Ability-based Trust (6 items, $\alpha$ = .96) (1 = Strongly disagree, 7 = Strongly agree)**
(1) The partner firm is very capable of performing its role in the alliance; (2) The partner firm is known to be successful at the things it tries to do; (3) The partner firm is well qualified for the alliance; (4) The partner firm has much knowledge about the work that needs to be done in the alliance; (5) We are very confident about the partner firm's skills; (6) The partner firm has specialized capabilities that add value to the alliance.

**Benevolence-based Trust (5 items, $\alpha$ = .92) (1 = Strongly disagree, 7 = Strongly agree)**
(1) While making important decisions, the partner firm is concerned about our company's welfare; (2) The partner firm would not knowingly do anything to hurt our company; (3) Our firm's needs are important to the partner firm; (4) The partner firm looks out for what is important to our firm in the alliance; (5) The partner firm will go out of its way to help our firm.

**Integrity-based Partner Trust (6 items, $\alpha$ = .94) (1 = Strongly disagree, 7 = Strongly agree)**
(1) The partner firm has a strong sense of justice; (2) The partner firm is fair in business dealings with us; (3) This alliance partner stands by its word; (4) The partner firm's behaviors are not very consistent (reverse coded item); (5) We like the partner firm's values and ideals; (6) Sound principles seem to guide the partner firm's actions.

**Mutual Power or Influence (3 items, $\alpha$ = .83) (1 = Strongly disagree, 7 = Strongly agree)**
(1) Our firm can influence the partner firm to change its decisions regarding R&D, sales, production, or distribution; (2) Partner firm can influence our firm to change the decisions regarding R&D, sales, production, or distribution [the difference between scales (1) and (2) are reverse-coded to measure power equality]; (3) Our firm and the partner company have equal say in all the business dealings in the alliance; (4) Our firm and the partner firm have equal influence on each other on all alliance-related decisions.

**Formal Control (4 items, $\alpha$ = .78) (1 = Strongly disagree, 7 = Strongly agree)**
(1) We often consult with legal experts to sort out the problems in the alliance; (2) We strictly follow the contracts to coordinate this alliance; (3) We rely on legal means to ensure that the partner firm meets its obligations; (4) Rules have been strictly enforced in the alliance.

**Interfirm Learning (4 items, $\alpha$ = .91) (1 = not at all, 7 = a great deal)**
(1) Our firm has learned to jointly execute marketing, R&D, or production operations with this alliance partner; (2) Our firm has learned to exchange skills, know-how, or technologies with the partner company; (3) Our firm has gained new techniques, competencies or technologies from this partner; (4) Our firm has developed new ideas or skills because of the strategic alliance with this partner.

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## References


Ahuja, Gautam
2000 'Collaboration networks, structural holes, and innovation: A longitudinal study'. *Administrative Science Quarterly* 45: 426–455.

Alter, Catherine, and Jerard Hage
1993 *Organizations working together*. Beverly Hills, CA: Sage.

Argote, Linda, Sara L. Beckman, and Dennis Epple
1990 'The persistence and transfer of learning in industrial settings'. *Management Science* 36: 140–154.

Argyris, Chris, and Donald A. Schön
1978 *Organizational learning: A theory of action perspective*. Reading, MA: Addison-Wesley.

Ariño, Africa, and Yves Doz
2000 'Rescuing troubled alliances: Before it's too late'. *European Management Journal* 18/2: 173–182.

Ariño, Africa, and Jose de la Torre
1998 'Learning from failure: Towards an evolutionary model of collaborative ventures'. *Organization Science* 9: 306–325.

Bagozzi, R. P., and L. W. Phillips
1982 'Representing and testing organizational theories: A holistic construal'. *Administrative Science Quarterly* 27: 459–489.

Blau, Peter M.
1964 *Exchange and power in social life*. New York: Wiley.

Bleeke, Joel, and David Ernst
1993 *Collaborating to compete: Using strategic alliances and acquisitions in the global marketplace*. New York: Wiley.

Bromiley, Philip, and Larry L. Cummings
1992 'Transaction costs in organizations with trust'. Working paper No. 28, Strategic Management Research Center, University of Minnesota, Minneapolis.

Browning, Larry D., Janice M. Beyer, and Judy C. Shetler
1995 'Building cooperation in competitive industry: SEMATECH and the Semiconductor industry'. *Academy of Management Journal* 38: 113–151.

Campbell, Donald T., and Donald W. Fiske
1959 'Convergent and discriminant validation by the multitrait-multimethod matrix'. *Psychological Bulletin* 56: 81–105.

Cohen, Wesley M., and Daniel A. Levinthal
1990 'Absorptive capacity: A new perspective on learning and innovation'. *Administrative Science Quarterly* 35: 128–152.

Cyert, Richard M., and James G. March
1963 *A Behavioral theory of the firm*. Englewood Cliffs, NJ: Prentice Hall.

Das, T. K., and Bing-Sheng Teng
1998 'Between trust and control: Developing confidence in partner cooperation in alliances'. *Academy of Management Review* 23/3: 491–512.

Dillman, Don A.
1978 *Mail and telephone surveys: The total design method*. New York: Wiley.

Dore, Ronald P.
1987 *Taking Japan seriously*. Stanford, CA: Stanford University Press.

Doz, Yves
1996 'The evolution of cooperation in strategic alliances: Initial conditions or learning processes?'. *Strategic Management Journal* 17: 55–83.

Dwyer, Robert F., Paul H. Schurr, and Sejo Oh
1987 'Developing buyer–seller relationships'. *Journal of Marketing* 51 (April): 11–27.

Dyer, Jeffrey H., and Harbir Singh
1998 'The relational view: Competitive strategy and sources of interorganizational competitive advantage'. *Academy of Management Review* 23/4: 660–679.

Geringer, Michael J., and Louis Hebert
1991 'Measuring performance of international joint ventures'. *Journal of International Business Studies* 22: 249–264.

Granovetter, Mark
1985 'Economic action and social structure: A theory of embeddedness'. *American Journal of Sociology* 91/3: 481–510.

Gulati, Ranjay
1995 'Does familiarity breed trust? The implications of repeated ties for contractual choice in alliances'. *Academy of Management Journal* 38: 85–112.

Gulati, Ranjay, Nitin Nohria, and Akbar Zaheer
2000 'Strategic networks'. *Strategic Management Journal* 21/3: 203–217.

Gundlach, Gregory T., Ravi S. Achrol, and John T. Mentzer
1995 'The structure of commitment in Exchange'. *Journal of Marketing* 59/1: 78–92.

Hagedoorn, John, and Jos Schakenraad
1994 'The effect of strategic technology alliances on company performance'. *Strategic Management Journal* 15/4: 291–310.

Hamel, Gary
1991 'Competition for competence and inter-partner learning within international strategic alliances'. *Strategic Management Journal*, (Summer Special Issue) 12: 83–103.

Harman, H. H.
1967 *Modern factor analysis*. Chicago: The University of Chicago Press.

Hedberg, Bo. L. T., Paul C. Nystrom, and William H. Starbuck
1976 'Camping on seesaws: Prescriptions for a self-designing organization'. *Administrative Science Quarterly* 21: 41–65.

Hill, Charles W. L.
1990 'Cooperation, opportunism, and the invisible hand: Implications for transaction cost theory'. *Academy of Management Review* 15: 500–513.

Homans, George C.
1961 *Social behavior*. New York: Harcourt, Brace and World.

Homans, G. C.
1976 'Commentary' in *Advances in Experimental Social Psychology*. L. Berkowitz and E. Walster (eds), 9: 231–244.

Hosmer, Larue Tone
1995 'Trust: The connecting link between organizational theory and philosophical ethics'. *Academy of Management Review* 20: 379–403.

Inkpen, Andrew C.
1998 'Learning and knowledge acquisition through international strategic alliances'. *Academy of Management Executive* 12/4: 69–80.

James, Larry R., William F. Joyce, and John W. Slocum
1988 'Organizations do not cognize'. *Academy of Management Review* 13: 129–132.

John, George
1984 'An empirical investigation of some antecedents of opportunism in a marketing channel'. *Journal of Marketing Research* 21 (August): 278–289.

Kabanoff, B.
1991 'Equity, equality, power, and conflict'. *Academy of Management Review* 16: 416–441.

Khanna, Tarun, Ranjay Gulati, and Nitin Nohria
1998 'The dynamics of learning alliances: Competition, cooperation, and relative scope'. *Strategic Management Journal* 19: 193–210.

Killing, J. Peter
1983 *Strategies for joint venture success*. New York: Praeger.

Kogut, Bruce
1988 'Joint ventures: Theoretical and empirical perspectives'. *Strategic Management Journal* 9/4: 319–332.

Konrad, Alison M., and Frank Linnehan
1995 'Formalized HRM structures: Coordinating equal employment opportunity or concealing organizational practices?' *Academy of Management Journal* 38: 787–820.

Kumar, Nirmalya, Louis W. Stern, and James C. Anderson
1993 'Conducting interorganizational research using key informants'. *Academy of Management Journal* 36/6: 1633–1651.

Lane, Christel, and Reinhard Bachmann
2000 *Trust within and between organizations*. Oxford: Oxford University Press.

Lane, Peter J., and Michael H. Lubatkin
1998 'Relative absorptive capacity and interorganizational learning'. *Strategic Management Journal* 19/5: 461–477.

Lazaric, Nathalie
1998 'Trust and organizational learning during inter-firm cooperation' in *Trust and Economic Learning*. N. Lazaric and E. Lorenz (eds), 209–226. Cheltenham: Edward Elgar.

Leenders, Roger T. A. J., and Shaul M. Gabbay
1999 *Corporate social capital and liability*. Boston, MA: Kluwer.

Lyles, Marjorie A., and Jane E. Salk
1996 'Knowledge acquisition from foreign parents in international joint ventures: An empirical examination in the Hungarian context'. *Journal of International Business Studies* 27/5: 877–903.

Macneil, Ian R.
1980 *The new social contract*. New Haven, CT: Yale University Press.

March, James G., and Herbert A. Simon
1958 *Organizations*. New York: Wiley.

Mayer, Roger C., and James H. Davis
1999 'The effect of the performance appraisal system on trust for management: A field of quasi-experiment'. *Journal of Applied Psychology* 84/1: 123–136.

Mayer, Roger C., James H. Davis, and David F. Schoorman
1995 'An integrative model of organizational trust'. *Academy of Management Review* 20/3: 709–734.

Nooteboom, Bart
1996 'Toward a learning based model of transactions' in *Transaction Cost Economics and Beyond*. J. Groenewegen (ed.). Boston, MA: Kluwer.

Nooteboom, Bart
2000 'Institutions and forms of coordination in innovation systems'. *Organization Studies* 21/5: 915–939.

Nunnally, Jum C., and Ira H. Bernstein
1994 *Psychometric theory*. New York: McGraw-Hill.

Parkhe, Arvind
1993 'Strategic alliance structuring: A game theoretic and transaction cost examination of interfirm cooperation'. *Academy of Management Journal* 36/4: 794–829.

Pfeffer, Jeffrey, and Gerald R. Salanick
1978 *The external control of organizations*. New York: Harper & Row.

Pisano, Gary P.
1989 'Using equity participation to support exchange: Exchange from the biotechnology industry'. *Journal of Law, Economics, and Organization* 5/1: 109–126.

Podsakoff, Philip M., and Dennis W. Organ
1986 'Self-reports in organizational research: Problems and prospects'. *Journal of Management* 12: 531–544.

Polanyi, Michael
1948 *Personal knowledge: Towards a post-critical philosophy*. Chicago: The University of Chicago Press.

Powell, Walter W., Kenneth W. Koput, and Laurel Smith-Doerr
1996 'Interorganizational collaboration and the locus of innovation: Networks of Learning in biotechnology'. *Administrative Science Quarterly* 41: 116–145.

Provan, Keith G.
1984 'Interorganizational cooperation and decision making autonomy in a consortium multihospital system'. *Academy of Management Review* 9: 494–504.

Provan, Keith G., and Jule B. Gassenheimer
1994 'Supplier commitment in relational contract exchanges with buyers: A study of interorganizational dependence and exercised power'. *Journal of Management Studies* 31/1: 55–68.

Ring, Peter Smith, and Andrew H. Van de Ven
1992 'Structuring cooperative relationships between Organizations'. *Strategic Management Journal* 13: 483–498.

Ring, Peter Smith, and Andrew H. Van de Ven
1994 'Developmental processes of cooperative interorganizational relationships'. *Academy of Management Review* 19: 90–118.

Rotter, Julian B.
1967 'A new scale for the measurement of interpersonal trust'. *Journal of Personality* 35: 651–665.

Ruekert, Robert W., and Orville Walker, Jr
1987 'Marketing's interaction with other functional units: A conceptual framework and empirical evidence'. *Journal of Marketing* 51/1: 1–19.

Sako, Mari
2000 'Does trust improve business performance?' in *Trust Within and Between Organizations.* C. Lane and R. Bachmann (eds). Oxford: Oxford Press.

Simonin, Bernard L.
1997 'The importance of collaborative know-how: An empirical test of the learning organization'. *Academy of Management Journal* 40/5: 1150–1174.

Steensma, Kevin H., and Marjorie A. Lyles
2000 'Explaining IJV survival in a transitional economy through social exchange and knowledge-based perspective'. *Strategic Management Journal* 21: 831–851.

Tully, Shawn
1996 'Northwest and KLM: The alliance from Hell'. *Fortune*, 24 June, pp. 64–71.

Williamson, Oliver E.
1985 *The Economic institutions of capitalism: Firms, markets, relational contracting*. New York: Free Press.

Williamson, Oliver E.
1991 'Comparative economic organization: The analysis of discrete structural alternatives'. *Administrative Science Quarterly* 36: 269–296.

Yan, Aimin, and Barbara Gray
(1994) 'Bargaining power, management control, and performance in United States-China joint ventures: A comparative case study'. *Academy of Management Journal* 37: 1478–1517.

Yoshino, Michael Y., and U. Srinivasa Rangan
1995 *Strategic alliances: An entrepreneurial approach to globalization.* Boston, MA: Harvard Business School Press.

Zaheer, Akbar, and N. Venkatraman
1995 'Relational governance as an interorganizational strategy: An empirical test of the role of trust in economic exchange'. *Strategic Management Journal* 16: 373–392.

Zaheer, Akbar, Bill McEvily, and Vincenzo Perrone
1998 'Does trust matter? Exploring the effects of interorganizational and interpersonal trust on performance'. *Organization Science* 9/2: 141–159.

Zand, Dale E.
1972 'Trust and managerial problem solving'. *Administrative Science Quarterly* 17: 229–239.

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